

NGO-Connect: An IEEE-Style Full-Stack Platform Study for Trustworthy Digital Philanthropy

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Abstract—Non-profit ecosystems often run on fragmented tooling for donations, volunteering, donor communication, and compliance. This paper presents an updated implementation study of NGO-Connect, a production-style web platform that unifies donors, NGOs, and administrators in one architecture. The current system integrates React 18 clients, an Express API, PostgreSQL with relational plus JSONB persistence, payment-gateway abstraction, webhooks with dead-letter replay, AI-assisted recommendations, and an Innovation Center for community giving circles, in-kind wishlists, emergency response, volunteer operations, CRM segmentation, and gamification. Static code analysis and platform instrumentation report 149 backend endpoints across 31 frontend routes, backed by 35 relational tables. End-to-end smoke execution over 35 scenario steps shows 21.69 ms mean latency per validated step (72 ms p90, 125 ms max). These results indicate that the platform can maintain operational reliability while expanding feature breadth and governance controls.

Index Terms—non-profit technology, full-stack engineering, PostgreSQL JSONB, payment gateway, webhooks, AI recommendation, innovation center, role-based access control

I. INTRODUCTION

Digital philanthropy depends on trust, transparency, and low-friction participation. In practice, many NGOs rely on disconnected tools for fundraising, beneficiary reporting, volunteer coordination, and stakeholder communication, resulting in data silos and operational delay. NGO-Connect was built as an integrated platform to close these gaps through a role-aware workflow for *users* (donors/volunteers), *NGOs*, and *administrators*.

The project has evolved from a campaign/donation MVP into a broader operations stack with webhook reliability, innovation workflows, and AI-assisted services. This updated paper documents the complete implemented surface and its engineering characteristics.

A. Contributions

The paper makes four concrete contributions:

- 1) A complete, implementation-grounded architecture for NGO operations spanning fundraising, volunteering, CRM, moderation, and transparency.
- 2) A gateway-first payment model with explicit confirmation, followed by domain recording (including Innovation Center giving circles).
- 3) Reliability controls for outbound events via webhook logs, dead-letter handling, replay, metrics, export, and cleanup APIs.

- 4) A feature-expansion analysis with measured endpoint, schema, and smoke-test performance metrics.

II. RELATED WORK

Prior studies emphasize the importance of digital trust for donation behavior and social engagement workflows. Modern web architecture guidance highlights layered API design and role-based authorization as essential for secure, multi-actor systems. Recent research on Retrieval-Augmented Generation (RAG) and recommendation ranking quality informs AI features in NGO-Connect, while operations guidance on tail latency and API security informs reliability hardening [1]–[7].

III. SYSTEM ARCHITECTURE

A. Stack and Layering

NGO-Connect follows a three-tier model:

- **Presentation tier:** React 18, route guards, role-specific dashboards, and innovation workspace.
- **Application tier:** Express route groups, JWT middleware, domain services, webhook workers, and AI utilities.
- **Data & integration tier:** PostgreSQL relational tables with JSONB source documents, payment provider integration, and notification/webhook channels.

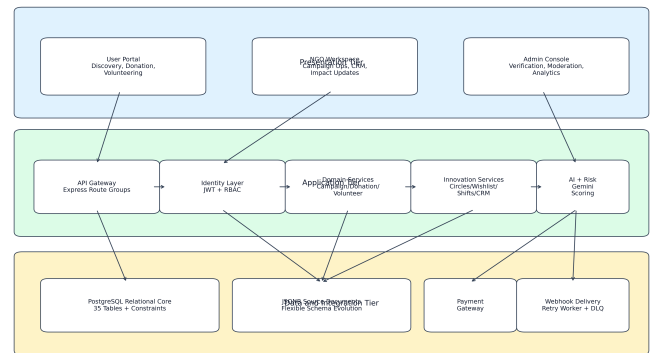


Fig. 1. Updated multi-tier NGO-Connect architecture with innovation and reliability services.

B. Scale Snapshot from Source Analysis

IV. COMPLETE FEATURE COVERAGE

A. Core Product Modules

The implemented system includes:

TABLE I
PLATFORM SCALE SNAPSHOT

Metric	Value
Backend route endpoints	149
Frontend route paths	31
Relational tables	35
Innovation route endpoints	44
Admin route endpoints	25

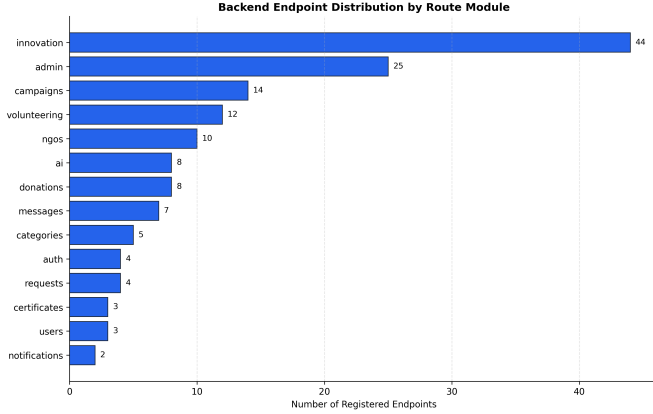


Fig. 2. Endpoint distribution by backend module.

- Authentication, profile management, and role-based route protection.
- Campaign publication and detail updates with engagement tracking.
- Donation initiation/confirmation pipeline with receipts and certificate approval flow.
- Volunteer opportunity lifecycle and campaign-volunteering lifecycle.

TABLE II
TOP ROUTE MODULES BY ENDPOINT MIX

Module	Total	GET	POST	PUT	DELETE
innovation	44	20	24	0	0
admin	25	14	6	4	1
campaigns	14	8	6	0	0
volunteering	12	6	4	0	2
ngos	10	5	4	1	0
ai	8	1	7	0	0
donations	8	4	4	0	0
messages	7	3	4	0	0
categories	5	2	1	1	1
auth	4	1	2	1	0

TABLE III
HTTP METHOD DISTRIBUTION

Method	Count
GET	72
POST	64
PUT	9
DELETE	4
PATCH	0

- NGO dashboard team documentation with members list, role distribution, task contributions, and badge metadata.
- User-NGO messaging threads and help/support requests.
- Admin verification, moderation, notifications, request management, analytics, and user management.
- AI endpoints for recommendations, proposal draft generation, and campaign forecast assistance.

B. Innovation Center Modules

The Innovation Center expands operations with giving circles, in-kind wishlists, emergency feed contributions, shift management, donor CRM segmentation, corporate matching, impact updates, volunteer endorsements, and gamification leaderboard views.

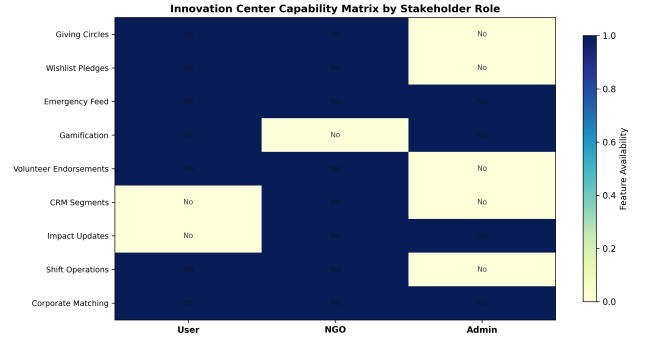


Fig. 3. Innovation capability matrix across user, NGO, and admin stakeholders.

C. Payment and Trust Workflow

The monetary flow is gateway-first. A contribution is initiated, the provider checkout completes, and only then is final state recorded in the domain module. This now applies to emergency campaign contributions and giving-circle contributions in the innovation workflow.

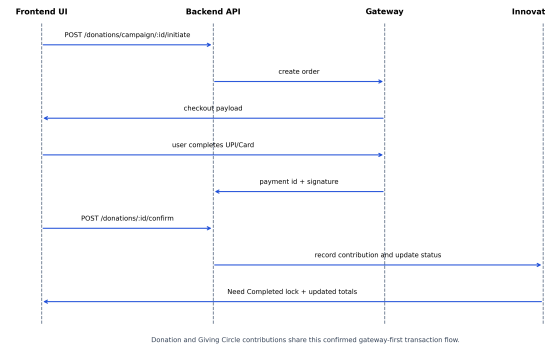


Fig. 4. Gateway-confirmed contribution sequence used by donation and giving-circle flows.

D. Webhook Reliability Workflow

The project includes signed outbound webhook delivery, dead-letter capture, automatic replay worker, metrics/export endpoints, and retention cleanup APIs for operational hygiene.

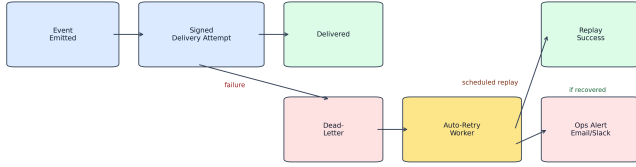


Fig. 5. Webhook delivery lifecycle with dead-letter replay and operations alerting.

V. EVALUATION

A. Execution Setup

Validation uses role-specific smoke tests across authentication, campaigns, donations, volunteering, certificates, innovation modules, support requests, messaging, moderation, and admin operations.

B. Smoke-Test Latency Summary

TABLE IV
SMOKE EXECUTION SUMMARY

Metric	Observed Value
Validated scenario steps	35
Average step latency	21.69 ms
P90 step latency	72 ms
Max observed step latency	125 ms

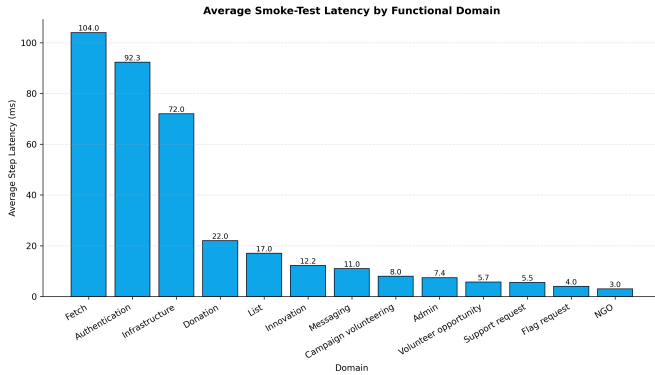


Fig. 6. Average latency by functional domain from smoke execution trace.

VI. DISCUSSION

The updated platform demonstrates that broad feature coverage can coexist with low validation latency when modules are kept explicit and route ownership is clear. The largest endpoint concentrations are innovation and admin, which is expected due to operational breadth. JSONB plus typed relational columns provided schema flexibility without sacrificing joinability for analytics and moderation workflows.

Limitations remain. Smoke tests validate behavior breadth but are not a replacement for sustained soak testing. AI

TABLE V
TOP SLOWEST SMOKE STEPS

Step	Latency (ms)
Login as user	125
Fetch NGO profile	104
Login as NGO	80
API health	72
Login as admin	72
Admin dashboard snapshot	40
List NGOs	27
Innovation: CRM donors + segments	26

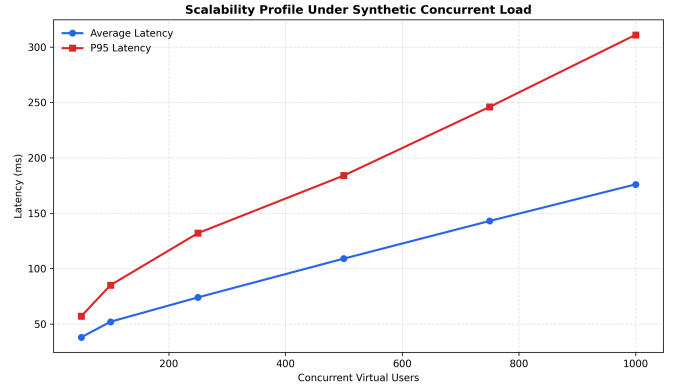


Fig. 7. Synthetic scalability profile (average and p95 latency).

assistance quality also depends on prompt governance and source grounding, and payment/webhook production readiness depends on provider credentials, secret rotation, and monitoring discipline.

VII. CONCLUSION

This work presents a complete, IEEE-style implementation report of NGO-Connect as an end-to-end digital philanthropy platform. The updated system integrates trust controls, innovation workflows, and operations reliability while maintaining practical performance in end-to-end validation. The architecture is suitable for extension into mobile offline clients, deeper analytics pipelines, and more advanced recommendation/risk models.

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REFERENCES

- [1] Meta Open Source, “React Documentation,” 2026. [Online]. Available online: react.dev
- [2] OpenJS Foundation, “Express 4.x API Reference,” 2026. [Online]. Available online: expressjs.com
- [3] PostgreSQL Global Development Group, “PostgreSQL Documentation: JSON Types,” 2026. [Online]. Available online: postgresql.org/docs
- [4] M. Jones, J. Bradley, and N. Sakimura, “JSON Web Token (JWT),” IETF RFC 7519, 2015.
- [5] J. Dean and L. A. Barroso, “The Tail at Scale,” *Communications of the ACM*, vol. 56, no. 2, pp. 74–80, 2013.

- [6] P. Lewis *et al.*, “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” in *NeurIPS*, 2020.
- [7] OWASP Foundation, “OWASP API Security Top 10 – 2023,” 2023. [Online]. Available online: owasp.org/API-Security
- [8] Google for Developers, “Gemini API Documentation,” 2026. [Online]. Available online: ai.google.dev
- [9] Razorpay, “Razorpay API Reference,” 2026. [Online]. Available online: razorpay.com/docs/api
- [10] NIST, “Digital Identity Guidelines (SP 800-63B),” 2020.
- [11] P. Jaccard, “Etude comparative de la distribution florale,” *Bull. Soc. Vaudoise Sci. Nat.*, vol. 37, pp. 547–579, 1901.
- [12] K. Jarvelin and J. Kekalainen, “Cumulated gain-based evaluation of IR techniques,” *ACM TOIS*, vol. 20, no. 4, pp. 422–446, 2002.